Portfolio Project #01

Global Firepower 2025 Analysis – A Strategic Insight

Business Questions and Insights

Q1: Does higher defense spending always lead to stronger military power?

Top 30 countries by defense expenditure

	country	defense_budget_billion	power_index	overall_power_rank
0	United States	895.00	0.0744	1
1	China	266.85	0.0788	3
2	Russia	126.00	0.0788	2
3	India	75.00	0.1184	4
4	Saudi Arabia	74.76	0.4201	24
5	United Kingdom	71.50	0.1785	6
6	Japan	57.00	0.1839	8
7	Australia	55.70	0.3298	18
8	France	55.00	0.1878	7
9	Ukraine	53.70	0.3755	20
10	Germany	50.00	0.2601	14
11	Poland	48.70	0.3776	21
12	Turkiye	47.00	0.1902	9
13	South Korea	46.30	0.1656	5
14	Canada	41.00	0.5179	28
15	Italy	30.89	0.2164	10
16	Israel	30.50	0.2661	15
17	Netherlands	27.00	0.6412	36
18	Brazil	26.16	0.2415	11
19	Switzerland	25.04	0.7869	44
20	Algeria	25.00	0.3589	26
21	Spain	23.40	0.3242	17
22	Taiwan	19.74	0.3988	22
23	Paraguay	17.60	1.9044	87
24	Iran	15.45	0.3048	16
25	Singapore	15.00	0.5271	29
26	Morocco	13.40	1.1273	59
27	Sweden	13.00	0.4835	27
28	Indonesia	10.60	0.2557	13
29	Colombia	10.54	0.8353	46

While extremely high defense spending enables top-tier military capability, increased budgets do not guarantee proportional improvements in power. After a threshold, strategic allocation, institutional efficiency, and technological integration play a greater role than raw expenditure.

Insight

High defense spending does not automatically translate into higher military power.

- Some countries with very large budgets ranked lower than expected
- Several countries achieved strong rankings with moderate spending

Business Meaning

How your defense budget is allocated matters more than how much is spent

Q2: Are countries efficient in converting budget into power?

Top 30 countries by budget efficiency

	country	defense_budget_billion	overall_power_rank	budget_rank	efficiency_gap
0	United States	895.00	1	1.0	0.0
1	China	266.85	3	2.0	-1.0
2	Russia	126.00	2	3.0	1.0
3	India	75.00	4	4.0	0.0
4	Saudi Arabia	74.76	24	5.0	-19.0
5	United Kingdom	71.50	6	6.0	0.0
6	Japan	57.00	8	7.0	-1.0
7	Australia	55.70	18	8.0	-10.0
8	France	55.00	7	9.0	2.0
9	Ukraine	53.70	20	10.0	-10.0
10	Germany	50.00	14	11.0	-3.0
11	Poland	48.70	21	12.0	-9.0
12	Turkiye	47.00	9	13.0	4.0
13	South Korea	46.30	5	14.0	9.0
14	Canada	41.00	28	15.0	-13.0
15	Italy	30.89	10	16.0	6.0
16	Israel	30.50	15	17.0	2.0
17	Netherlands	27.00	36	18.0	-18.0
18	Brazil	26.16	11	19.0	8.0
19	Switzerland	25.04	44	20.0	-24.0
20	Algeria	25.00	26	21.0	-5.0
21	Spain	23.40	17	22.0	5.0
22	Taiwan	19.74	22	23.0	1.0
23	Paraguay	17.60	87	24.0	-63.0
24	Iran	15.45	16	25.0	9.0
25	Singapore	15.00	29	26.0	-3.0
26	Morocco	13.40	59	27.0	-32.0
27	Sweden	13.00	27	28.0	1.0
28	Indonesia	10.60	13	29.0	16.0
29	Colombia	10.54	46	30.0	-16.0

Defense spending efficiency varies significantly across countries. Several mid-sized nations outperform their budget ranks through strategic allocation and institutional effectiveness, while some high-spending countries underperform due to structural and operational inefficiencies. Top military powers combine scale with disciplined spending.

Insight:

Military efficiency varies significantly across countries

- Some countries outperformed their budget rank
- Others underperformed despite high spending

Business Meaning

Strategic planning and institutional effectiveness drive military outcomes.

Q3: Manpower vs Technology — which matters more?

Top 15 countries by weapons technology

	country	active_personnel	overall_power_rank	total_aircraft	aircraft_per_100k_personnel
0	United States	1328000.0	1	13043.0	982.15
1	United Arab Emirates	65000.0	54	551.0	847.69
2	Bahrain	18000.0	81	132.0	733.33
3	Sweden	24000.0	27	169.0	704.17
4	Finland	24000.0	48	163.0	679.17
5	Denmark	20000.0	45	117.0	585.00
6	Japan	247000.0	8	1443.0	584.21
7	Australia	57000.0	18	327.0	573.68
8	Slovenia	7000.0	96	39.0	557.14
9	Montenegro	2000.0	127	11.0	550.00
10	New Zealand	9000.0	86	48.0	533.33
11	Austria	16000.0	68	85.0	531.25
12	Canada	68000.0	28	351.0	516.18
13	Zambia	15000.0	109	77.0	513.33
14	France	200000.0	7	976.0	488.00

Countries with limited manpower tend to adopt technology-intensive military strategies, using advanced airpower to compensate for smaller forces. While major powers also maintain high technology density, manpower-heavy countries generally show lower technology intensity.

This indicates that modern military effectiveness increasingly depends on technological leverage rather than sheer troop size.

Insight:

Technology acts as a force multiplier in modern warfare.

- Smaller militaries with high aircraft-per-personnel ratios ranked strongly
- Large manpower-heavy forces underperformed without matching technology

Business Meaning

Quality increasingly outweighs quantity in military capability.

Q4. Logistics vs Weapons — what matters more?

Top 15 countries with strong logistics

	country	total_aircraft	logistics_score	overall_power_rank
0	United States	13043.0	1.000000	1
1	China	3309.0	0.358249	3
2	India	2229.0	0.323594	4
3	Brazil	513.0	0.195698	11
4	Canada	351.0	0.172820	28
5	Russia	4292.0	0.160907	2
6	Japan	1443.0	0.135098	8
7	Australia	327.0	0.122979	18
8	United Kingdom	631.0	0.114800	6
9	France	976.0	0.099328	7
10	Germany	584.0	0.091375	14
11	Italy	729.0	0.090433	10
12	Mexico	433.0	0.083199	32
13	Indonesia	459.0	0.080048	13
14	Sweden	169.0	0.068777	27

While weapons strength is essential for combat capability, logistics infrastructure is the key enabler of sustained military power. Countries with strong logistics consistently achieve high rankings even with moderate weapon counts, whereas weapons-heavy countries with weak logistics struggle to project power effectively.

Insight

Logistics is a critical enabler of military effectiveness.

- Top-ranked countries had strong logistics capacity
- Weapons-heavy countries with weak logistics ranked lower

Business Meaning

Operational sustainability matters as much as combat power

Q5: Geography & Resources — do they determine power?

Top 15 countries with strong geography

	country	geography_score	resource_score	overall_power_rank
0	Russia	0.796025	0.263322	2
1	Canada	0.496549	0.560548	28
2	China	0.476151	0.085646	3
3	Brazil	0.436061	0.041868	11
4	United States	0.402551	0.125769	1
5	Kazakhstan	0.296449	0.098746	57
6	India	0.246870	0.015174	4
7	Argentina	0.207019	0.008163	33
8	Democratic Republic of the Congo	0.187774	0.000592	66
9	Indonesia	0.181358	0.008163	13
10	Vietnam	0.176005	0.014483	23
11	Colombia	0.155497	0.006715	46
12	Australia	0.149958	0.008064	18
13	Chile	0.135983	0.000494	47
14	Peru	0.121983	0.002831	49

Geography is a foundational enabler of military power that sets upper limits on strategic potential. However, only countries that combine geographic advantages with strong logistics, technology, and industrial capacity are able to convert this potential into sustained dominance which is evident from the rankings above.

Top 15 countries with strong resources

	country	geography_score	resource_score	overall_power_rank
0	Venezuela	0.092836	1.000000	50
1	Saudi Arabia	0.082255	0.851190	24
2	Iran	0.094815	0.686613	16
3	Canada	0.496549	0.560548	28
4	Iraq	0.061823	0.477338	43
5	Kuwait	0.006166	0.334090	79
6	United Arab Emirates	0.014720	0.321912	54
7	Russia	0.796025	0.263322	2
8	Libya	0.076220	0.159178	76
9	United States	0.402551	0.125769	1
10	Nigeria	0.085480	0.121425	31
11	Kazakhstan	0.296449	0.098746	57
12	China	0.476151	0.085646	3
13	Brazil	0.436061	0.041868	11
14	Algeria	0.111024	0.040157	26

Natural resource abundance supports long-term sustainability and defense funding, but it does not translate directly into military power. Resource-rich countries frequently underperform militarily due to weak industrial integration, limited technological development, and governance constraints. The strongest militaries rely more on industrial capacity and innovation than on raw resource ownership.

Insight:

Geography and natural resources create potential, not dominance.

- Resource-rich countries often ranked mid or low
- Large geographic countries underperformed without capability investment
- Top powers had moderate resources but strong integration

Business Meaning

Structural advantages must be converted into operational capability.

Project Summary:

This project analyses global military power using a multi-dimensional, data-driven approach. By combining financial, manpower, technological, logistical, geographic, and natural resource indicators, the study evaluates how different strategic factors contribute to overall military effectiveness.

Key findings show that extremely high defense spending enables top-tier power but exhibits diminishing returns beyond a threshold. Budget efficiency varies widely, with several mid-sized nations outperforming their spending levels. Technology intensity increasingly outweighs raw manpower, while logistics infrastructure emerges as the strongest enabler of sustained military capability. Geographic scale and natural resources provide structural advantages but do not independently guarantee dominance without strong institutional and industrial integration.

Overall, the analysis demonstrates that modern military strength is driven by balanced investment, efficient resource utilization, and strong operational infrastructure rather than reliance on any single factor.